

Condensed Novelty Check: Delay–Layout Coupling in UWB Self-Calibration

Executive summary for paper planning

1 Thesis

UWB anchor self-calibration can trade antenna-delay parameters against layout scale, so same-environment tag accuracy is not a reliable proxy for physical calibration quality. In the Erlangen campaign, V4 is metrically distorted (Sim3 scale 0.958) and V5 is metric-correct (1.010), yet V4 barely wins under scalar p50 static LOO positioning (61.5 vs 63.5 mm, a 2.0 mm gap); after reducing static tag-anchor positive range bias, V5 wins (44.5 vs 53.9 mm). This ranking flip is the key evidence that the wrong geometry can win by cancelling structured NLOS bias, because the intervention effect is proportionally larger than the baseline gap.

2 Literature Gap

What is known	What is NOT known (our gap)
UWB self-calibration estimates anchors from radio ranges, robot motion, multihop links, or auxiliary sensing [5, 3, 6]. Antenna-delay and range-bias calibration are established UWB problems [7, 15, 12, 8]. NLOS mitigation uses channel statistics, learned models, robust weighting, or biased-range models [14, 4]. Fisher/CRLB and rigidity analyses describe weak directions and rank deficiencies in localization networks [13, 1]. Scale-propagation-parameter ambiguity is known in acoustic self-calibration and related ranging systems [2]. AniTrack observed that self-localized anchors outperformed surveyed anchors in tag accuracy [10].	These papers do not test whether the layout with the best tag accuracy is metrically correct. Delay is treated as a nuisance correction, not as a parameter that can hide global layout-scale error. NLOS reduction is not used as a causal intervention to reveal which calibration geometry is physically valid. Prior work does not connect a weak delay–layout direction to a measured wrong-geometry-wins result in UWB self-calibration. The interaction with structured UWB NLOS bias, and the resulting ranking flip between distorted and metric-correct layouts, is not reported. It reports the paradox but does not diagnose Sim3 scale, delay–layout coupling, NLOS cancellation, or profile/Fisher structure.

3 Our Contribution

The first contribution is scientific: we quantitatively characterize delay–layout–NLOS coupling in UWB anchor self-calibration. The evidence chain combines the V4/V5 Sim3 split, Fisher information with weakest eigenvalue 10^{-6} , profile-likelihood and morph-valley surfaces, phase-center robustness, and the raw-frame lower-tail intervention. The decisive result is the ranking flip: V4 barely wins when p50 tag-anchor positive bias is present, but V5 wins after that bias is reduced, showing that V4’s p50 advantage came from cancellation rather than better physical calibration.

The second contribution is methodological: we provide a falsification protocol that separates metric correctness from same-environment positioning accuracy. It uses Sim3 anchor-scale validation, a layout/delay transfer matrix, winner’s-curse correction, nested spatial CV, phase-center sensitivity, NLOS leakage auditing, and ROTO dynamic checks to prevent a single accuracy number from hiding the mechanism.

The same failure mode can occur beyond UWB whenever per-device bias and geometric scale share a near-degenerate Fisher direction, including GNSS clock–geometry coupling, acoustic sound-speed–geometry coupling, and other range-only calibration systems.

4 Nearest Prior Art

Paper	What they did	How we differ
Hamer and D’Andrea 2018 [5]	Built a self-calibrating one-way UWB network with clock synchronization and anchor computation for multi-robot localization.	We test whether the empirically best radio-only layout is physically correct, and show that a distorted layout can win by cancelling NLOS bias.
Lutz et al. 2019 [9]	Used visual-inertial SLAM and fiducial markers to estimate UWB anchor poses and sensor error-model parameters.	Their calibration is aided by external visual structure; ours diagnoses a range-only self-calibration degeneracy using Vicon only as independent validation.
Ledergerber and D’Andrea 2018 [7]	Estimated module-dependent range inaccuracies by maximum likelihood without external ground truth.	We show that fitting delay/range inaccuracies can reward a geometrically wrong solution unless anchor metric scale is checked separately.
Luder et al. 2025 (AniTrack) [10]	Reported that self-localized DWM3000 anchors gave better tag accuracy than laser-surveyed anchors (13.96 vs 16.57 cm average error).	They report an unexplained paradox; we provide the mechanism, Sim3/Fisher/profile evidence, and the NLOS-reduction intervention.
Pan et al. 2015 (GNSS) [11]	Showed satellite-clock estimation can absorb orbit error and improve PPP positioning accuracy.	Their coupled-error absorption is a GNSS clock/orbit effect; our result is a UWB self-calibrated-geometry error that makes the physically wrong layout beat the correct one.

5 Risk and Status

Risk	Severity	One-line mitigation
One room and 24 static positions	High	State that transfer remains unproven and freeze the pipeline for second-room validation.
Cross-domain scale–delay ambiguity is prior art	Medium	Claim quantitative UWB characterization plus NLOS-induced ranking flip, not the bare ambiguity.
Lower-tail result is static batch, not real-time	Medium	Present it as a diagnostic intervention and static engineering candidate, not a deployable NLOS classifier.
High-error tail remains	Medium	Separate median mechanism evidence from P95/dynamic performance claims.
Phase-center uncertainty	Low	Use sensitivity analysis and avoid treating Vicon-oracle rank as the sole proof.

Status: 129 papers indexed, 15 cited, 0 explain delay–layout–NLOS coupling. Target: *Measurement* (IF ~ 5.6). Title: “When Wrong Geometry Wins: Delay–Layout Coupling in UWB Self-Calibration.”

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